Ved Shah

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Education

University of Maryland Aug. 2026 – May 2030
B.S. Computer Engineering

Downingtown STEM Academy Class of 2026
GPA: 4.45/4.0 weighted | SAT: 1540 | National Merit Finalist

Experience

Research Assistant May – July 2025
Penn State University, CENSAI Lab
- Researched arithmetic in LLaMA 3.2-3B using interpretability tools to trace intermediate calculations and operator precedence (PEMDAS); developed an embedding-swap technique to alter precedence behavior.
- Authored a research paper accepted to CoLM and presented findings at the conference.

Research Assistant June – Aug. 2025
University of Delaware, Sensify Lab
- Developed Vocal Goals, a speech improvement platform that visualizes voice qualities for practice, with a focus on gender-affirming voice training.
- Used Python, Librosa, and OpenSMILE to analyze and visualize pitch, resonance, formants, and other voice characteristics.

Robotics Researcher July – Aug. 2025
Partners in Science Program, Liberty Science Center
- Designed a multi-robot warehouse inventory simulation using ROS, Gazebo, and deep reinforcement learning, integrating global task allocation with local robot decision-making.
- Conducted research during an 8-week program at Columbia Engineering and presented at a symposium.

Software Engineering Intern May – Aug. 2024
The Community Grocer, 501(c)(3)
- Engineered a web app to streamline ordering for SNAP recipients, coordinate community events, and schedule meal deliveries, increasing access to healthy food options.

Projects

Preventa AI | *Co-founder, Software Developer* Aug. – Sept. 2025
- Built PreventaAI, a healthcare app with medication tracking, an AI assistant, and a visual scanner; built Eyesight, an accessibility app using LiDAR and on-device CoreML for obstacle detection and real-time narration.

Slicer AI | *Founder, Software Developer* June – Nov. 2024
- Developed an AI tennis coaching app using 3D human pose estimation, cosine similarity, and 3D neural networks to correct tennis swings from video.
- Won 1st at Congressional App Challenge (top 0.1%), 2nd at Penn State LionCage, 4th at Husky Dog Pound, and top 10 at Diamond Challenge.

Fibromyalgia Diagnosis | *Researcher* Nov. – Jan. 2024
- Investigated ML diagnosis using HRV, GSR, and EMG features, achieving approximately 94% test accuracy; applied feature scaling and GAN augmentation, feeding GCN embeddings into a 2D CNN.
- Placed 3rd of 60+ in Computer Science at Delaware Valley Science Fairs and 1st at CCRSC fair (2025).

Competitions and Hackathons

Foundry, "Startup in a Weekend" — 3rd place: Built GlassBoxAI to detect AI-assisted interview answers live over Zoom and Google Meet using eye tracking, facial landmarks, and voice analysis.

NexHacks: Built Kinetix, an AI physical therapy and athletic training coach on Meta Quest using real-time computer vision, joint-angle biomechanics, and voice feedback. Demo

HackUMBC — 1st in Game Design: Built His Royal Flyness, a co-op game using a simulated fruit fly brain, Godot, WebSocket, Python, and Supabase. Demo

Technical Skills

Languages: Python, Java, Swift, JavaScript
Frameworks and Databases: React, Next.js, Node/Express, Flask, Supabase, MongoDB, PostgreSQL, Redis
Tools and Concepts: AWS, Docker, CI/CD, Git, GitHub, VS Code, Arduino, Godot, REST APIs, Data Structures, OOP, UX